

Task 3:

Name of the aircraft:

Passenger capacity: 299

Engine count: 2 engines

Detailed calculation:

According to task 1, the total trip time required is 102.48813844421308, which is approximately 102. Let passenger capacity is x .

(1) When $250 \leq x < 300$, which requires 2 engines. Flight time is 10, 11 or 12 times. The minimum flight time is 10 when $x=299$

$$\text{Trip fuel} = 20 \times 2 \times 102$$

$$\text{Time related cost per minute of flight (Ct)} = 22$$

$$C = 0.76 \times 40 \times 102 \times 102 + 22 \times 102 + 2000 = 320525.6$$

$$C' = C \times 10 = 3205256$$

(2) $300 \leq x < 350$, which requires 4 engines. Flight time is at least 9 times.

$$\text{Trip fuel} = 20 \times 4 \times 102$$

$$\text{Time related cost per minute of flight (Ct)} = 24$$

$$C = 0.76 \times 80 \times 102 \times 102 + 24 \times 102 + 2500 = 637511.2$$

$$C' = C \times 9 = 5737600.8$$

(3) $350 \leq x < 400$, which also requires 4 engines. Flight time is at least 8 times.

$$\text{Time related cost per minute of flight (Ct)} = 26$$

$$C = 0.76 \times 80 \times 102 \times 102 + 26 \times 102 + 2500 = 637715.2$$

$$C' = C \times 8 = 5101721.6$$

(4) $400 \leq x < 450$, which also requires 4 engines. Flight time is at least 7 times.

$$\text{Time related cost per minute of flight (Ct)} = 28$$

$$C = 0.76 \times 80 \times 102 \times 102 + 28 \times 102 + 2500 = 637919.2$$

$$C' = C \times 7 = 4465434.4$$

Therefore, when the passenger is 299, the quantity of C is the least, which means it achieves minimum cost.